

A disproof of Conjecture 00000000223

The density of $\{n : q \mid E_n\}$ is 0, not a value of q^{-2} type

1 The conjecture

Conjecture 00000000223 reads:

Definition: Euclid numbers $E_n = p_n\# + 1$ (one plus the product of the first n primes).

Conjecture: The least prime factor of E_n exceeds p_n uniformly for all sufficiently large n , and the least prime factors are uniformly distributed on the logarithmic scale over (p_n, p_n^2) ; moreover, for a fixed prime q , the density of n with q dividing E_n is an explicit value of q^{-2} type.

The Chinese text of the last clause says the density is a “ q^{-2} -type explicit value” (a value of q^{-2} form). The conjecture is the conjunction of three clauses: (A) least prime factor $> p_n$; (B) log-scale equidistribution of least prime factors; (C) the density clause. We refute (C), which refutes the conjunction. Clauses (A) and (B) are not addressed. (Clause (A) is in fact true by the same argument, and (B) is open; neither is used.)

2 Definitions and conventions

Following the conjecture and https://en.wikipedia.org/wiki/Euclid_number (“ $E_n = p_n\# + 1$, where $p_n\#$ is the n th primorial (the product of the first n prime numbers)”):

- $p_1 = 2, p_2 = 3, \dots$; in Lean $p_{i+1} = \text{Nat.nth Nat.Prime } i$.
- $p_n\# = \prod_{i < n} \text{nth Prime } i$ (Lean `primorialFirst`); Mathlib’s `primorial` is the product of the primes $\leq n$, a different function, and is not used.
- $E_n = p_n\# + 1$ (Lean `euclid`); $D_q = \{n \in \mathbb{N} : q \mid E_n\}$ (Lean `divSet`).
- $S \subseteq \mathbb{N}$ has natural density d if $\#(S \cap [0, N])/N \rightarrow d$ (Lean `HasNatDensity`). We use $0 \in \mathbb{N}$; $E_0 = 2$. Adding or removing the single index $n = 0$ does not change any density.

3 Readings of “an explicit value of q^{-2} type”

We refute every reading that requires the density to be nonzero for at least one prime q , in particular:

- (i) exact form: the density equals cq^{-2} for every prime q , for some constant $c \neq 0$;
- (ii) asymptotic form: the densities $d(q)$ exist and $q^2 d(q) \rightarrow c \neq 0$ as $q \rightarrow \infty$ through the primes (this is $d(q) \sim cq^{-2}$, and covers explicit values such as $1/(q^2 - 1)$).

Not refuted: a pure upper-bound reading “the density is $O(q^{-2})$ ”. The value 0 satisfies it. We do not regard it as the literal reading: the text asks for an explicit *value* of q^{-2} type (form), not a bound, and 0 has no dependence on q at all. We state this reading explicitly so that the reviewer can weigh it.

4 The proof

Lemma 1 (Euclid; Lean `not_dvd_euclid`). *If $r < n$ then $p_{r+1} \nmid E_n$.*

Proof. p_{r+1} is one of the factors of $p_n\#$. If it also divided $p_n\# + 1$ it would divide 1; but it is prime. (Compare https://en.wikipedia.org/wiki/Euclid%27s_theorem: “If p divides P and q , then p must also divide the difference of the two numbers, which is $(P + 1) - P$ or just 1.”) $\square$

Lemma 2 (Lean `divSet_subset`, `hasNatDensity_zero_of_subset`). *For a prime q let $r = \#\{\text{primes} < q\}$, so $q = p_{r+1}$ (Mathlib `Nat.nth_count`). Then $D_q \subseteq [0, r]$, hence $\#(D_q \cap [0, N)) \leq r + 1$ and D_q has natural density 0.*

Proof. The inclusion is the previous lemma. Then $0 \leq \#(D_q \cap [0, N))/N \leq (r + 1)/N \rightarrow 0$. $\square$

Theorem 3 (Lean `conjecture_223_density_clause_false`).

1. *For every prime q , D_q has natural density 0 (`hasNatDensity_divSet`).*
2. *There is no prime q and $d \neq 0$ such that D_q has density d (`no_nonzero_density`).*
3. *Reading (i) is false: there is no $c \neq 0$ with density c/q^2 for all primes q (`not_exact_form`; it already fails at $q = 2$).*
4. *Reading (ii) is false: there are no $d(\cdot)$ and $c \neq 0$ with D_q of density $d(q)$ for all primes q and $p_{r+1}^2 d(p_{r+1}) \rightarrow c$ as $r \rightarrow \infty$ (`not_asymptotic_form`).*

Proof. (1) is the lemma. Limits in $\mathbb{R}$ are unique, so any density of D_q equals 0; this gives (2), and (3) at $q = 2$. For (4), $p_{r+1}^2 d(p_{r+1}) = 0$ for all r , so the limit is $0 \neq c$. $\square$

Example: $E_6 = 30031 = 59 \cdot 509$, so $59 \mid E_6$, but $59 = p_{17}$ divides no E_n with $n \geq 17$.

5 Lean formalization

File `lean/Conjecture223/Basic.lean` (129 lines; only import `Mathlib`). Main theorem `C223.conjecture_223_density_clause_false`, statement as in the Theorem above. Build: `lake build` succeeds; `#print axioms` for the capstone and for `no_nonzero_density`, `not_exact_form`, `not_asymptotic_form` reports only `propext`, `Classical.choice`, `Quot.sound`. No sorry, no `native_decide`, no new axioms.